

Password Storage in Databases

From Naive to Secure

Why Does Password Storage Matter?

The Challenge:

- Users need to authenticate with passwords
- Systems must verify passwords
- Databases can be compromised

The Stakes:

-  Stolen credentials can affect multiple sites (password reuse)
-  Legal and regulatory compliance (GDPR, CCPA)
-  Company reputation and trust
-  Financial damages from breaches

Method 1: Plaintext Storage

 Never Do This

Plaintext Storage

How it works: Store passwords exactly as users enter them.

```
CREATE TABLE users (  
  id INT PRIMARY KEY,  
  username VARCHAR(50),  
  password VARCHAR(100) -- Stored as-is  
);  
  
INSERT INTO users VALUES (1, 'alice', 'MyPassword123');  
INSERT INTO users VALUES (2, 'bob', 'SecretPass456');
```

Login Check:

```
SELECT * FROM users  
WHERE username = 'alice' AND password = 'MyPassword123';
```

Plaintext Storage - Issues

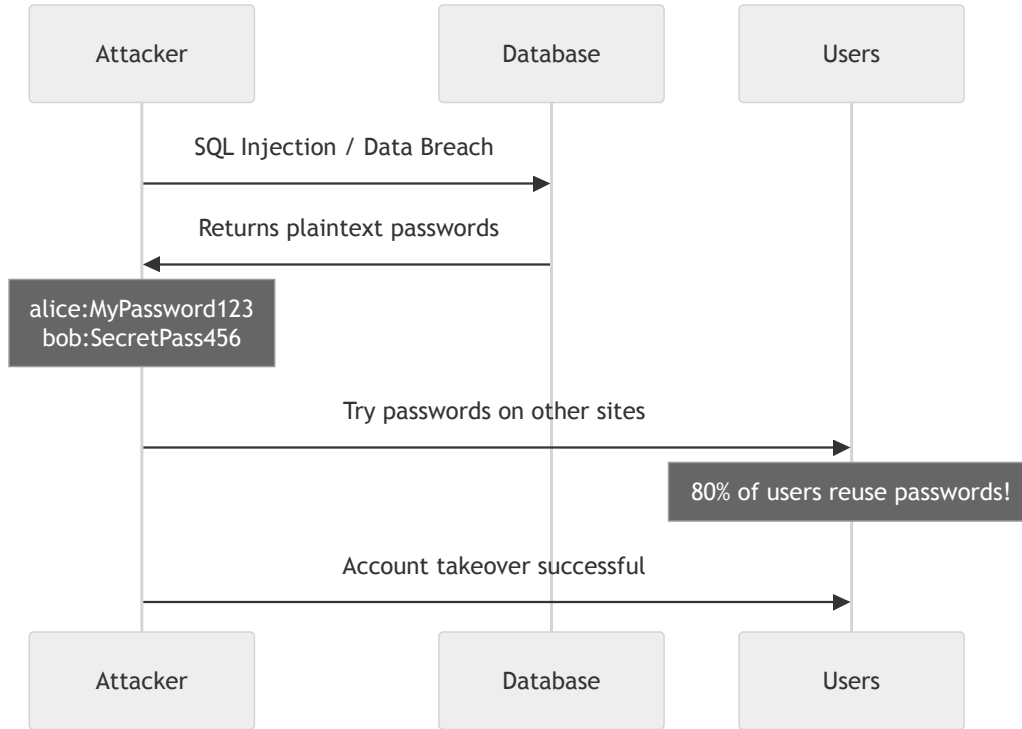
Critical Vulnerabilities:

Issue	Impact
💣 Database Breach	All passwords immediately exposed
💣 Insider Threat	DBAs and developers can see all passwords
💣 Logs & Backups	Passwords visible in logs, backups, debugging
💣 No Protection	Zero security if database is compromised
💣 Legal Risk	Violates data protection regulations

Real-World Example:

```
Database dump:  
alice | MyPassword123
```

Plaintext Storage - Attack Scenario



Verdict: **✗ Completely Unacceptable**

Method 2: Simple Encryption

● Still Not Good Enough

Simple Encryption

How it works: Encrypt passwords with a symmetric key before storing.

```
const crypto = require("crypto");

// Encryption
function encryptPassword(password, secretKey) {
  const cipher = crypto.createCipher("aes-256-cbc", secretKey);
  let encrypted = cipher.update(password, "utf8", "hex");
  encrypted += cipher.final("hex");
  return encrypted;
}

// Decryption
function decryptPassword(encrypted, secretKey) {
  const decipher = crypto.createDecipher("aes-256-cbc", secretKey);
  let decrypted = decipher.update(encrypted, "hex", "utf8");
  decrypted += decipher.final("utf8");
  return decrypted;
}
```

```
INSERT INTO users VALUES (1, 'alice', 'a3f5b8c2e1d4 ...');
```

Simple Encryption - Issues

Problems:

Issue	Impact
● Key Management	If encryption key is stolen, all passwords exposed
● Reversible	Passwords can be decrypted back to plaintext
● Key in Code	Encryption key often hardcoded or in config
● Single Point of Failure	One key protects everything
● Insider Threat	Anyone with key access can decrypt all passwords

Attack Scenario:

Attacker finds encryption key in:
- Source code repository

Method 3: Basic Hashing

🟡 Better, but Still Vulnerable

Basic Hashing (Without Salt)

How it works: Use a one-way hash function to store passwords.

```
const crypto = require("crypto");

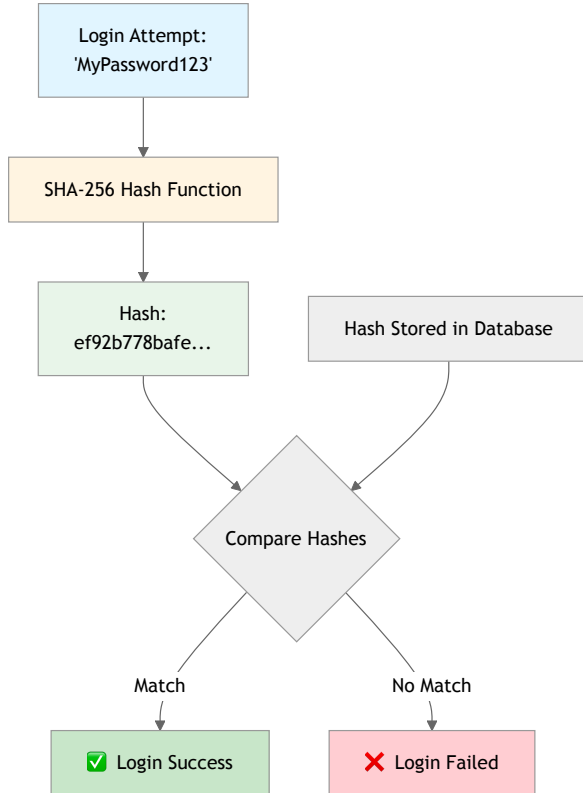
function hashPassword(password) {
  return crypto.createHash("sha256").update(password).digest("hex");
}

// Storing
const passwordHash = hashPassword("MyPassword123");
// Result: ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f
```

```
CREATE TABLE users (
  id INT PRIMARY KEY,
  username VARCHAR(50),
  password_hash VARCHAR(64)
);

INSERT INTO users VALUES (
  1,
  'alice',
  'ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f'
);
```

Basic Hashing - How It Works



Basic Hashing - Issues

Vulnerabilities:

Issue	Description
🟡 Rainbow Tables	Pre-computed tables of hash:password pairs
🟡 Identical Hashes	Same password = same hash across all users
🟡 Common Passwords	Attackers can quickly identify common passwords
🟡 Fast Computation	Modern GPUs can compute billions of hashes/second
🟡 No Uniqueness	No user-specific variation

Example Attack:

```
Database contains:  
alice | ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f
```

Rainbow Table Attack

What are Rainbow Tables? Pre-computed tables that map hashes back to passwords.

Sample Rainbow Table (SHA-256):

Password	SHA-256 Hash
password	5e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a11ef721d1542d8
123456	8d969eef6ecad3c29a3a629280e686cf0c3f5d5a86aff3ca12020c923adc6c92
Password123	ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f
qwerty	65e84be33532fb784c48129675f9eff3a682b27168c0ea744b2cf58ee02337c5
12345678	ef797c8118f02dfb649607dd5d3f8c7623048c9c063d532cc95c5ed7a898a64f
admin	8c6976e5b5410415bde908bd4dee15dfb167a9c873fc4bb8a81f6f2ab448a918
letmein	1c8bfe8f801d79745c4631d09ffff36c82aa37fc4cce4fc946683d7b336b63032
welcome	b2aa58a3c8807bb40c1d0da5e1df234143c3fed5abe6e8e4bf599ebfc2a36f37

How it works:

1. Attacker steals hash from database: ef92b778bafe ...
2. Looks up hash in rainbow table
3. Finds match: Password123
4. Can now login as that user!

Method 4: Hashing with Salt

🟢 Much Better!

Hashing with Salt

What is a Salt? A unique random value added to each password before hashing.

```
const crypto = require("crypto");

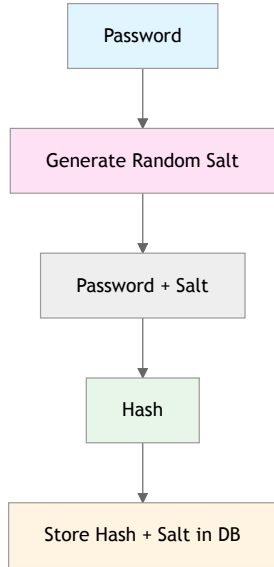
function hashPasswordWithSalt(password) {
  // Generate random salt (16 bytes = 128 bits)
  const salt = crypto.randomBytes(16).toString("hex");

  // Combine password and salt, then hash
  const hash = crypto
    .createHash("sha256")
    .update(password + salt)
    .digest("hex");

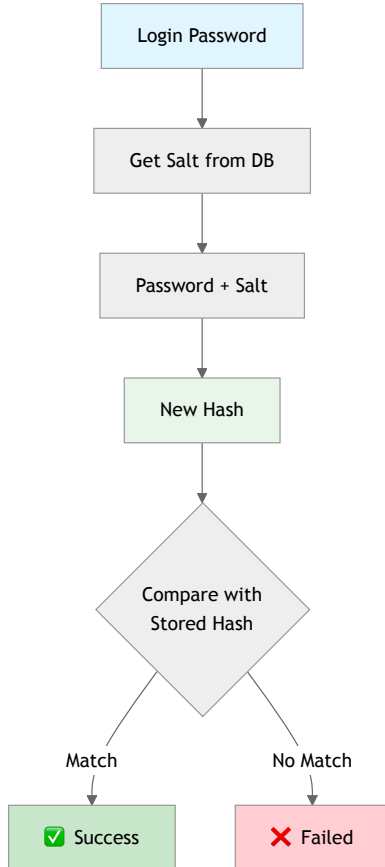
  // Return both hash and salt (need salt for verification)
  return { hash, salt };
}

// Store password
const result = hashPasswordWithSalt("MyPassword123");
console.log(result);
// {
//   hash: '7d8f9e2a1b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6b7c8d9',
//   salt: '4f7b3a9e2c1d8f6a'
// }
```

Salting Process - User Registration



Salting Process - Login Verification



Salt Benefits - Database View

Without Salt:

alice		ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f		NULL
bob		ef92b778bafe771e89245b89ecbc08a44a4e166c06659911881f383d4473e94f		NULL
carol		5e884898da28047151d0e56f8dc6292773603d0d6aabbdd62a11ef721d1542d8		NULL

⚠ Same password = same hash

With Salt:

alice		7d8f9e2a1b3c4d5e ...		4f7b3a9e2c1d8f6a
bob		a1b2c3d4e5f6a7b8 ...		9e8d7c6b5a4f3e2d
carol		3c4d5e6f7a8b9c0d ...		2f1e0d9c8b7a6f5e

✅ Same password = different hash (unique salts!)

Hashing with Salt - Benefits

Why Salts Work:

Benefit	Explanation
 Unique Hashes	Same password produces different hashes for each user
 Rainbow Table Defense	Pre-computed tables become useless
 Per-User Attack	Attacker must crack each password individually
 Increased Complexity	Each user requires separate brute-force attempt
 Pattern Breaking	Can't identify users with same password

Attack Impact:

Without salt: Crack 1 hash → Compromise all users with that password

With salt: Crack 1 hash → Compromise only 1 user

Method 5: Modern Password Hashing

● Industry Standard

Modern Password Hashing Algorithms

Purpose-Built for Passwords: These algorithms are intentionally slow and memory-intensive.

Algorithm	Year	Key Features
bcrypt	1999	Adaptive cost factor, built-in salt
scrypt	2009	Memory-hard, resistant to hardware attacks
Argon2	2015	Winner of Password Hashing Competition

Key Principles:

-  **Slow by Design:** Takes ~100-500ms to hash
-  **Memory Hard:** Requires significant RAM (scrypt, Argon2)
-  **Adjustable Cost:** Can increase difficulty over time
-  **Built-in Salt:** Salt generation and storage included
-  **Brute-Force Resistant:** Makes attacks computationally expensive

bcrypt Example

Using bcrypt (Node.js):

```
const bcrypt = require("bcrypt");

// Hash a password
async function hashPassword(password) {
  const saltRounds = 12; // Cost factor (2^12 iterations)
  const hash = await bcrypt.hash(password, saltRounds);
  return hash;
}

// Verify a password
async function verifyPassword(password, hash) {
  const match = await bcrypt.compare(password, hash);
  return match;
}

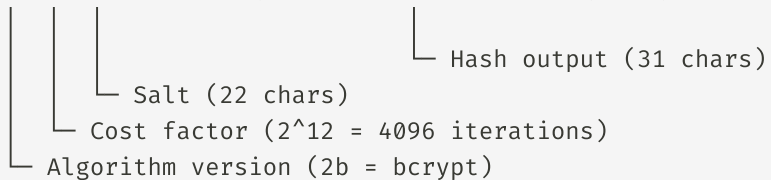
// Example usage
const password = "MyPassword123";
const hash = await hashPassword(password);
console.log(hash);
// $2b$12$KIXl5N4Y3Y6qx8N4LY0bF.eH4zKx2ZL1XYqGHLpQLcQ8z3xL4jY2W

// Verify
```

bcrypt Hash Anatomy

Understanding the bcrypt hash:

```
$2b$12$KIXl5N4Y3Y6qx8N4LY0bF.eH4zKx2ZL1XYqGHlPqLcQ8z3xL4jY2W
```



bcrypt Cost Factor Impact

Cost	Iterations	Time	Security
12	4,096	~250ms	Recommended default
14	16,384	~1s	High security
16	65,536	~4s	Very high security

⚡ Each +1 to cost = 2x slower = 2x more secure

Argon2 Example (Modern Winner)

Argon2 - The Current Gold Standard:

```
const argon2 = require("argon2");

// Hash a password
async function hashPassword(password) {
  const hash = await argon2.hash(password, {
    type: argon2.argon2id, // Hybrid mode (best of both worlds)
    memoryCost: 65536, // 64 MB (2^16 KiB)
    timeCost: 3, // 3 iterations
    parallelism: 4, // 4 parallel threads
  });
  return hash;
}

// Verify a password
async function verifyPassword(password, hash) {
  try {
    return await argon2.verify(hash, password);
  } catch (err) {
    return false;
  }
}
```


Attack Resistance Comparison

Cost to Crack **8-Character** Password:

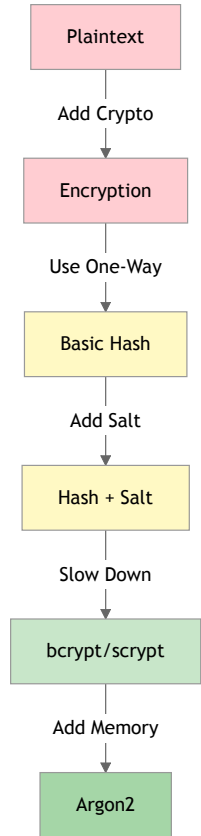
Hashing Method	Time to Crack	Security Level
 Plaintext	Instant	 None
 SHA-256 (No Salt)	Minutes	 Very Low
 SHA-256 (With Salt)	Hours	 Low
 bcrypt (cost=12)	Weeks	 High
 Argon2id (High Memory)	Months-Years	 Very High

Estimates based on modern GPU/ASIC hardware (2024)

Security Evolution Summary

Method	Security	Performance	Verdict
 Plaintext	None	Fast	Never use
 Encryption	Low	Fast	Wrong approach
 Basic Hash	Low-Medium	Fast	Vulnerable
 Hash + Salt	Medium	Fast	Better, but...
 bcrypt	High	Slow (good!)	Good choice
 Argon2	Very High	Slow (good!)	Best choice

The Evolution:



Real-World Password Breach Examples

LinkedIn (2012):

- 💣 6.5M passwords stolen
- ⚠️ Used unsalted SHA-1 hashes
- 📊 90% cracked within days

Adobe (2013):

- 💥 150M passwords stolen
- ⚠️ Used encrypted passwords with same key
- 📊 Weak encryption algorithm

LastPass (2022):

-  Used PBKDF2 with high iterations
-  Passwords not compromised
-  Strong hashing protected users

Lesson: Proper password hashing is crucial!

Key Takeaways

🎯 Never store passwords in plaintext or with reversible encryption

🎯 Always use purpose-built password hashing algorithms:

-  Argon2id (best choice)
-  bcrypt (solid choice)
-  scrypt (good alternative)

🎯 Defense in depth:

- Rate limiting
- Multi-factor authentication
- Strong password policies
- Regular security audits

Thank You!

Questions?